MAI Project AY 2026–27

ESE Presentation

Project ID: Group 12

Title of the Project: InfluenceIQ — Offline Decision-Discussion Detection with Source-Linked Meeting Review

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Introduction

“Where did we discuss the prototype decision?”

Find the passage

Relevant discussion can be scattered across the recording.

Know who spoke

Wrong speaker labels can credit the wrong participant.

Check the source

A candidate should lead back to its words and timestamp.

Problem framing: passage retrieval and evidence review. We do not assume any external product lacks these features.

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Problem Statement

Illustrative dialogue only — these are not measured model predictions.

Discussion

“Could we use the smaller prototype?
Let’s check its battery life first.”

Final-decision candidate

“We agree to use the smaller prototype.”

**The learned model suggests discussion passages.
Decision rules and human confirmation are separate.**

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Objectives of the Project

Process locally

Audio, speaker analysis and available slide content are processed on the laptop.

Keep evidence attached

Retain source text, segment IDs, speaker labels and recording timestamps.

Measure specific tasks

Compare discussion classifiers and speech components with matched references.

Required behaviour: record/upload → analyse → review → export

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Project Plan with Timeline

Phase sequence is documented; unverified month-by-month dates are not invented.

1 **Scope + literature**

2 **Prepare grouped data**

3 **Train + compare**

4 **Select combination**

5 **Integrate review**

6 **Evaluate + report**

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Literature Review

Research area	Established focus	Our implemented focus
CALO [1]	Integrated meeting assistance	Bound the task to local evidence review
Whisper / WhisperX [2–3]	Recognition and alignment	Use pretrained speech; no ASR novelty claim
Speakers [4–5, 7–8]	Speakers and multilingual evaluation	Measure speaker and language limits separately
Meeting datasets [10–11]	Summaries and supporting evidence	Provide extractive, source-linked suggestions

Integrated local review + a matched discussion-classifier comparison.

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Gap in the Research / Technology / Methodology

Established components exist; the project connects them into a reviewable local workflow.

Integration need

Connect words, speakers and optional slide content while retaining source IDs and timestamps.

Evaluation need

Measure discussion membership separately from speech errors and final-decision correctness.

Evidence still needed

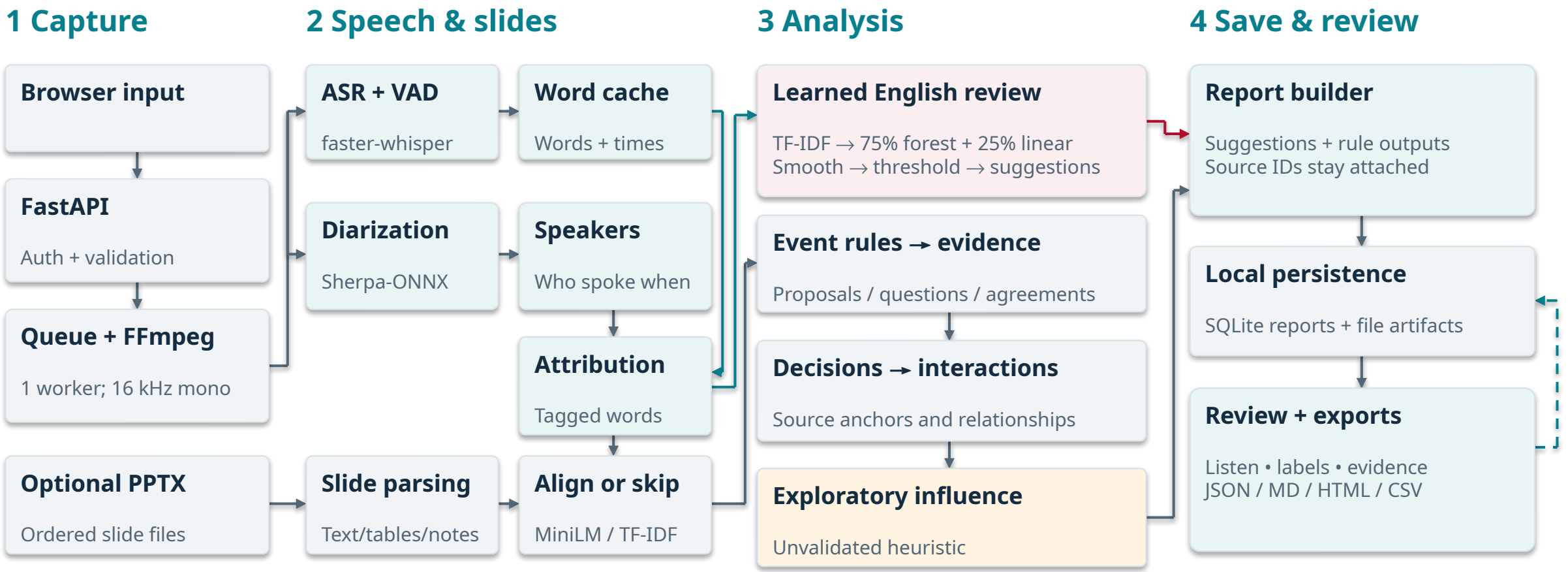
Validate complete recordings, slide links and human influence rankings on new annotations.

Implemented focus: local, source-linked discussion review.

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Description of the Proposed Solution

Data flow, model boundaries, persistence and human review — all on the laptop.



Arrows show data dependencies, not parallel compute. ASR drafts precede final speaker attribution; cached words are reused.

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Requirement Analysis

Functional requirements

Record or upload audio.
Optionally add presentation slides.
Review source text and timestamps.
Correct speaker display names.
Export source-linked reports.

Operational requirements

Preprovision local model assets.
Show explicit processing stages.
Queue one analysis job at a time.
Persist job and report state.
Keep optional-stage failures visible.

Speaker renaming changes display metadata; correcting acoustic clusters requires speaker analysis to be rerun.

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Technology Stack

Layer	Technology	Purpose
Interface	React + browser audio	Record, upload and source review
Local service	FastAPI + Python	Validation, queue and orchestration
Audio and speakers	FFmpeg / faster-whisper / Sherpa-ONNX	Normalize, recognise and diarize
Slides	PPTX parsing / MiniLM / TF-IDF	Extract text and propose links
Learned discussion	scikit-learn: TF-IDF + forest / logistic	English discussion suggestions
Storage and reports	SQLite + local artifacts	Saved review and report exports

Speech and embedding models are pretrained. Small supervised discussion classifiers are trained on the training partition.

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Design: Discussion Detector

Two conventional classifiers; no generative meeting summary is produced.

Reference acts

Human text + discussion labels

→

Lexical features

Words / characters + nearby context

→

Two scores

Random forest + logistic regression

→

Review output

Smooth scores + apply threshold

Combined score = 75% forest + 25% logistic

Context: ±40 acts. Smoothing: ±5 acts. Threshold: 0.357740. English-only learned review.

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Development / Implementation

47 AMI meetings with manual discussion spans; series remain in one partition.

Partition	Meetings	Acts	Positive acts
Training	29	20,339	5,862
Development	6	3,739	1,117
Historical	12	11,136	2,481

Compare 4 classifier families × 2 feature settings.
Choose settings on development data; retain all results.

Historical data has been examined across rounds: exploratory evidence, with only 3 independent historical project series.

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Testing & Debugging

597 recorded software tests

Checks include batching, source IDs, meeting boundaries, optional-stage errors and report/export integration.

Keep failed experiments visible

Semantic and continuity searches did not replace the baseline. Calibration and refit gains failed transfer checks.

597-test regression recorded 19 October 2026; it is not a count of correctly analysed meetings. Full negative results remain in the research report.

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Project Outcome: Discussion Detection

Historical cohort: human reference transcripts and act boundaries — not recording-level app accuracy.



39.94%

Precision

About 40 of 100 flagged acts match reference positives.

81.18%

Recall

Finds about 81 of 100 reference-positive acts.

Historical F1 (%) — common context-40 features

Mixture F1 53.54%; matched forest 52.08%. Difference +1.46 points; descriptive series-bootstrap interval −7.16 to +5.99.

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Project Outcome: Speech and Speakers

Word error rate: edits per 100 reference words, pooled by reference count. Lower is better.

	small model		large-v3-turbo
Hindi • 8 samples	50.96%	→	**28.85%**
Hinglish • 8 samples	66.13%	→	**57.26%**
English historical • 2 clips	61.61%	→	**63.91%**

Speaker pilot: DER 64.08% → 40.91%; speaker-count MAE 8.5 → 1.5.

Small stays the shared default; Turbo is selectable. Tiny fixed pilots, strict scoring and reference-coverage limits apply.

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Conclusion

Local processing
Source-linked review
Measured, bounded claims

A working prototype, a reproducible classifier comparison, and a clear plan for independent recording-level validation.

Still unmeasured: final-decision, slide-link and human influence accuracy. Fresh recording-level evaluation is needed.

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References

[1] Tur et al. (2010)
The CALO meeting assistant system.

[2] Radford et al. (2023)
Robust speech recognition via large-scale weak supervision.

[3] Bain et al. (2023)
WhisperX: Time-accurate speech transcription of long-form audio.

[4] Bredin (2023)
pyannote.audio 2.1 speaker diarization pipeline: Principle, benchmark, and recipe.

[5] Plaquet & Bredin (2023)
Powerset multi-class cross entropy loss for neural speaker diarization.

[7] Baghel et al. (2023)
The DISPLACE challenge 2023.

[8] Kalluri et al. (2024)
The second DISPLACE challenge: DIarization of SPeaker and LAnguage in Conversational Environments.

[10] Hu et al. (2023)
MeetingBank: A benchmark dataset for meeting summarization.

[11] Kim et al. (2023)
ExplainMeetSum: A dataset for explainable meeting summarization aligned with human intent.

[12] Carletta et al. (2006)
The AMI meeting corpus: A pre-announcement.

Selected entries retain IEEE paper numbering. All 15 full references are in the paper and this slide's notes.

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Photograph of the Presentation

Add the actual presentation photograph after the session.

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